

MUHAMMAD YASIR

Process Systems Engineer | Process Simulation, Optimisation and Computational Modelling

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PROFILE

Process systems engineer with five years across process and systems engineering, freelance simulation consulting and independent research. That work produced three first-author journal papers, two of them in *Energy* and *Separation and Purification Technology*, all built on one method: custom genetic algorithms in MATLAB and Python coupled directly to Aspen Plus and Aspen HYSYS through the COM interface. Results include a 26% cut in LNG compressor power and a 65% cut in carbon capture cost. PEC-registered chemical engineer, now adding a machine learning layer on top of this rigorous simulation base for faster, physics-consistent modelling.

PUBLICATIONS

1. **Yasir, M.**, Haider, J., Saeed, S., Amanipour, M., Roskilly, T. (2026). Integrated rotating packed bed reactor design for post combustion carbon capture: Parametric analysis, optimization and economic viability. *Separation and Purification Technology*, 386, 136584. doi:10.1016/j.seppur.2025.136584
2. **Yasir, M.**, Saeed, S., Khan, A., Ali, I. (2025). A thermo-economic optimization of dual multicomponent refrigerant cycles: Minimizing operational expenditure (OPEX) in natural gas liquefaction. *Energy*, 334, 137562. doi:10.1016/j.energy.2025.137562
3. **Yasir, M.**, Saeed, S., Khan, A., Ali, I. (2026). Energy penalty mitigation through exhaust gas recirculation: A technical evaluation of post-combustion carbon capture integrated with CCGT plants. *International Journal of Engine Research*. doi:10.1177/14680874251415355
4. **Yasir, M.**, et al. An integrated layer-resolved heat transfer and thermodynamic model for boil-off gas prediction in LH₂ storage tanks. *Under review*.

PROFESSIONAL EXPERIENCE

Systems Engineer

2021 to Present

Islamabad, Pakistan

- Process and combustion simulation for engineering programmes in Aspen Plus, Aspen HYSYS, MATLAB and Python: model development, equipment sizing, sensitivity analysis and convergence troubleshooting in multi-recycle systems.
- Simulation and performance results feed engineering decisions and design reviews, working with multidisciplinary engineering teams.

Freelance Process Design and Simulation Consultant

2019 to 2021

- Delivered 20+ process design and simulation projects for independent clients, working solo from scoping to final report.
- Focus on gas turbine performance modelling and LNG process optimisation, delivered as Aspen HYSYS and Aspen Plus simulation files, technical reports and supporting analysis.
- Example: calculated the hydrogen flowrate needed to replace natural gas fuel in a gas turbine while holding output at 397 MW, working the combustion and energy balance from first principles.

RESEARCH EXPERIENCE

Liquid Hydrogen Storage: Layer-Resolved Boil-Off Gas Modelling and Economics

2026 to Present

Collaboration: Durham University, United Kingdom

- Built a layer-resolved multilayer insulation (MLI) heat transfer model in Python, resolving each insulation layer at every timestep instead of assuming a fixed effective conductivity, coupled to a two-phase boil-off thermodynamic framework and an economic module.
- Global sensitivity analysis across seven design variables: vacuum pressure dominant (NS 12.76), ahead of fill level (9.60) and MLI layer count (5.80).
- Boil-off fraction rose from under 10% to over 50% across a 90-day run as vacuum degraded from 5×10^{-4} Pa to 0.46 Pa. Storage-related levelised cost of hydrogen \$6.19 to \$6.88/kg; optimum fill level about 75%, MLI optimum 60 to 70 layers.

Carbon Capture Using Rotating Packed Bed Technology

2024 to 2025

Collaboration: Durham University, United Kingdom

- Full-scale process model and design study for post-combustion carbon capture with rotating packed bed (RPB) reactors.
- Continuous absorber-regenerator loop in Aspen Plus with custom rate-based mass transfer correlations, coupled to a MATLAB genetic algorithm through the COM interface. The RPB itself was modelled from scratch in MATLAB, since Aspen Plus has no native RPB block.
- Parametric study across MEA concentrations (30, 45, 60 wt%) and lean loadings (0.16, 0.24, 0.30).
- Optimisation cut total annualised cost from \$84M to \$60.99M per year and CO₂ capture cost from \$224.6 to \$77.88 per tonne (65%); absorber height down 90% and packing volume down 95% against conventional packed beds.

Natural Gas Liquefaction Process Enhancement

2024 to 2025

Collaboration: NUST, Pakistan; Prince Mohammad Bin Fahd University, Saudi Arabia

- Optimised a dual mixed refrigerant (DMR) cycle for LNG at 27 bar with a custom MATLAB genetic algorithm linked to Aspen HYSYS through COM, querying the rigorous simulator across 12 decision variables.
- Compressor power down 26% (530.5 to 392.4 kJ/kg LNG); OPEX down 23% (\$14.61 to \$11.23 per tonne LNG).
- Sensitivity analysis showed a small subset of refrigerant composition variables dominated the objective function.

Energy Penalty Mitigation in CCGT-Integrated Carbon Capture

2023 to 2024

- Integrated Aspen Plus model of a full CCGT with post-combustion capture: gas turbine, HRSG, steam turbine and MEA absorption unit.
- Optimal exhaust gas recirculation ratio (0.33) raised flue gas CO₂ from 5.44% to 8.22% and cut reboiler heat duty by 34.5%, lowering steam cost from \$14.23 to \$9.26 per tonne CO₂ with net power penalty under 0.5%.

TECHNICAL SKILLS

Process Simulation	Aspen Plus and Aspen HYSYS (5+ years): rigorous modelling, EO mode, rate-based absorption, LNG and cryogenic systems, equipment sizing, sensitivity studies, convergence troubleshooting in multi-recycle flowsheets.
Optimisation	Custom genetic algorithms in MATLAB and Python driving Aspen Plus and HYSYS through COM automation, querying the rigorous simulator at every evaluation; multi-variable thermo-economic optimisation.
Programming and ML	MATLAB (advanced): algorithm development, numerical methods, COM automation. Python (intermediate): NumPy, SciPy, pandas, matplotlib, scikit-learn. PyTorch (developing): surrogate models for process systems.
Thermodynamics	Peng-Robinson, SRK and MBWR equations of state; NRTL; REFPROP; phase equilibria; refrigeration and liquefaction cycle design.
Heat and Mass Transfer	Heat exchanger network design and pinch analysis; shell-and-tube and plate exchangers; cryogenic MLI thermal modelling; packed, tray and RPB column design; atmospheric and vacuum distillation.
Economic Analysis	Global sensitivity analysis; techno-economic assessment (TEA); CAPEX, OPEX and levelised cost modelling.
Domain	Natural gas liquefaction (DMR, MR cycles); liquid hydrogen storage and boil-off; post-combustion carbon capture (MEA, RPB); combustion and power cycle simulation; process intensification.

EDUCATION

BSc Chemical Engineering

Sep 2015 to Aug 2019

NFC Institute of Engineering and Technology, Multan, Pakistan

- CGPA 3.56/4.00, top 10 of cohort. 4-year HEC-accredited programme, honours equivalent.
- Highest score in cohort (98/100) on the 100-question comprehensive exit examination.
- Final year project: heat integration in crude oil distillation, covering atmospheric and vacuum column design, pinch analysis and heat exchanger network optimisation.

REGISTRATION, CERTIFICATION AND LANGUAGES

Pakistan Engineering Council (PEC), Registered Chemical Engineer. Material Characterization and Identification, NCNDT, SES.
English: IELTS 7.0 (valid to January 2028). Urdu: native.